Sina Qasempour

Software Engineer · Backend & Data Systems · Open to research

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Software engineer building backend and data systems since 2020, from independent projects to municipal revenue platforms and an enterprise product I lead. Author of SageQL, an open-source text-to-SQL SDK in which the language model never writes SQL. I enjoy real research problems and am looking for research opportunities, with a strong interest in quantum computing.

EXPERIENCE

Zaravand Computer Co. — Software Developer

Full-stack development, later data and AI work, on the company's municipal and enterprise products.

Rahtal — enterprise work and performance management

2024 – present

Project lead · Data scientist · Full-stack developer

- Lead the project day to day: work with the product owner, own the overall design, and run the meetings that decide which features to build and how.
- Design the AI features planned for the product, including SageQL (below) and an in-product agent whose first version, an API integration companion, is public.
- Work as the project's data scientist and as a full-stack developer (Django/DRF, React).

Zaraamad — municipal revenue and service platform

2023 – 2026

Full-stack developer · Django, FastAPI, PostgreSQL, SQL Server

- Built the trade-duty subsystems for union and non-union businesses, the transferred-income duty subsystem, and the general "Other" subsystem, which accepts any type of municipal duty, calculates it and issues the bill to the citizen; roughly three quarters of the commits in each are mine.
- Built search across very large volumes of citizens' bills in these subsystems (trade, other-duty and payment-date search, person-name filtering).
- Also major contributor to the burial-services subsystem and the shared core (about 970 commits in the main backend, June 2023 – September 2026).

Zarvand and Tireban

Frontend developer

- Zarvand: citizen-services interface for accounts, municipal bills, requests and notifications (React, TypeScript).
- Tireban: web interface for tire condition and maintenance history across heavy mining-vehicle fleets.

Danobin — Chief Technology Officer

Startup · concurrent with Zaravand

- Set technical direction for the product and implemented its core services: subscription-plan changes and credit management, domain-event documentation, and an email service with templates, asynchronous tasks and tests (Django, Next.js).

Independent software projects

From 2020

Self-directed projects that built my foundation in software and led to my role at Zaravand.

RESEARCH AND OPEN SOURCE

Classical vs. Quantum-Kernel SVMs for Breast Cancer Classification

Sep 2026

Independent research · submitted to Machine Learning: Science and Technology (IOP Publishing); not accepted

- Controlled benchmark of tuned classical SVMs against quantum-kernel SVMs (ZZ feature maps, exact statevector simulation) on the Wisconsin Diagnostic Breast Cancer dataset, with nested cross-validation and fold-specific preprocessing.
- Found no quantum advantage: classical baselines achieved higher F1 on every outer split, quantum kernels degraded with circuit depth, and simulation was 45–80× slower. Repository

SageQL — text-to-SQL without model-written SQL

Sep 2026 – present

Open-source Python SDK · PyPI · live showcase

- A language model only extracts intent; deterministic code authorizes the request against the host's registered catalog and row policies and compiles parameterized, read-only SQL.
- Raised correct reports on a multi-turn benchmark from 8/27 to 22/27 while cutting model calls by 55%, by compiling complete intents locally instead of using a planner model; about 2,200 offline tests.

Vue.js documentation (Persian translation)

Apr 2024

Translated the testing guide; reviewed and merged as pull request #185.

EDUCATION

Bachelor's completion program, Professional Computer Engineering (Software)

Oct 2025 – Jul 2027 (expected)

Islamic Azad University, Kerman Branch

GPA 93.95 / 100 (18.79 / 20). Selected courses: Discrete Mathematics 100/100, Advanced Programming 100/100, Software Engineering Laboratory 100/100.

Associate degree, Computer

Chamran, Kerman

SKILLS

Languages & frameworks: Python, TypeScript/JavaScript, Django, Django REST Framework, FastAPI, NestJS, React, Next.js
Data: SQL, PostgreSQL, SQL Server, SQLAlchemy, Alembic, Pandas, NumPy, scikit-learn, Qiskit
Engineering: API and service integration, database design, testing, Docker, Linux, Git

CERTIFICATES AND LANGUAGES

IBM Data Science Professional Certificate (Jul 2026), including Machine Learning with Python and the Applied Data Science Capstone · **Foundations of Real World Math**, Saylor Academy (Sep 2026) · **Algebra: Elementary to Advanced**, Johns Hopkins University (Oct 2025). Full list: LinkedIn.

English: EF SET English Certificate 72/100, C2 Proficient (Reading 71, Listening 72), Jul 2026 · certificate. **Persian:** native.